Telecommunication Systems
ICE 3109

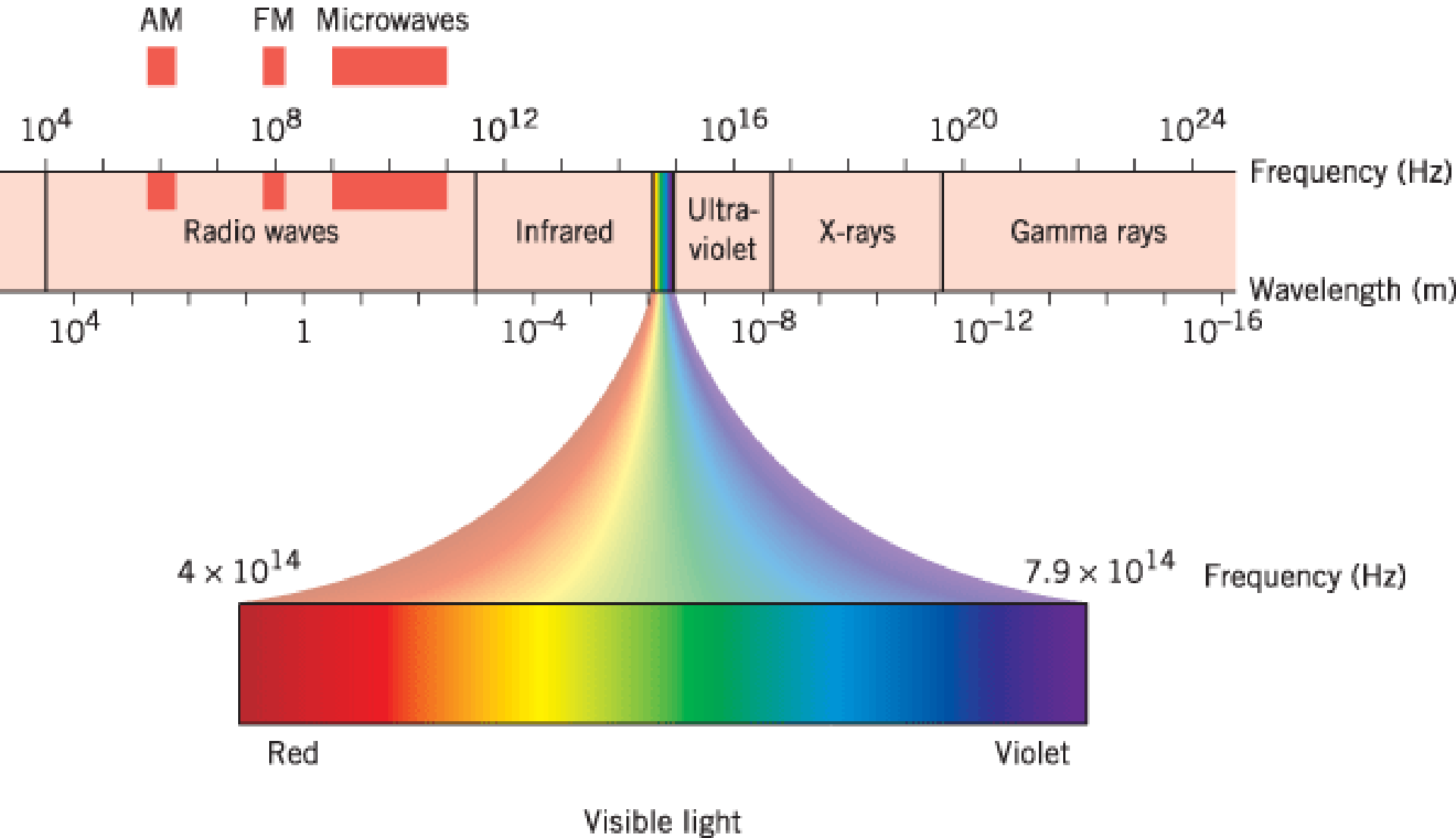
Lecture 7
OPTICAL FIBER COMMUNICATION

Introduction

- A method of transmitting information from one place to another by sending light through an optical fiber.

- The light forms an electromagnetic carrier wave that is modulated to carry information.

Electromagnetic spectrum



Evaluation of Fiber

- 1880 – Alexander Graham Bell
- 1930 – Patents on tubing
- 1950 – Patent for two-layer glass wave-guide
- 1960 – Laser first used as light source
- 1965 – High loss of light discovered
- 1970s – Refining of manufacturing process
- 1980s – OF technology becomes backbone of long distance telephone networks in NA.
- 1988s – ANSI developed SONET
- 1990s – Optical voice and data network world wide

Advantages of Optical Fiber

- Wider bandwidth and greater capacity
- Immunity to cross talk
- Immunity to static interference
- Environmental Immunity
- Safety and convenience
- Lower transmission loss
- Security
- Durability
- Reliability
- Economics
- Flexibility



Disadvantages of Optical Fiber

- Interfacing costs
- Strength
- Remote electrical power
- Susceptible to losses introduced by bending cable
- Specialized tools equipments and training
- Adding additional nodes

Areas of Application

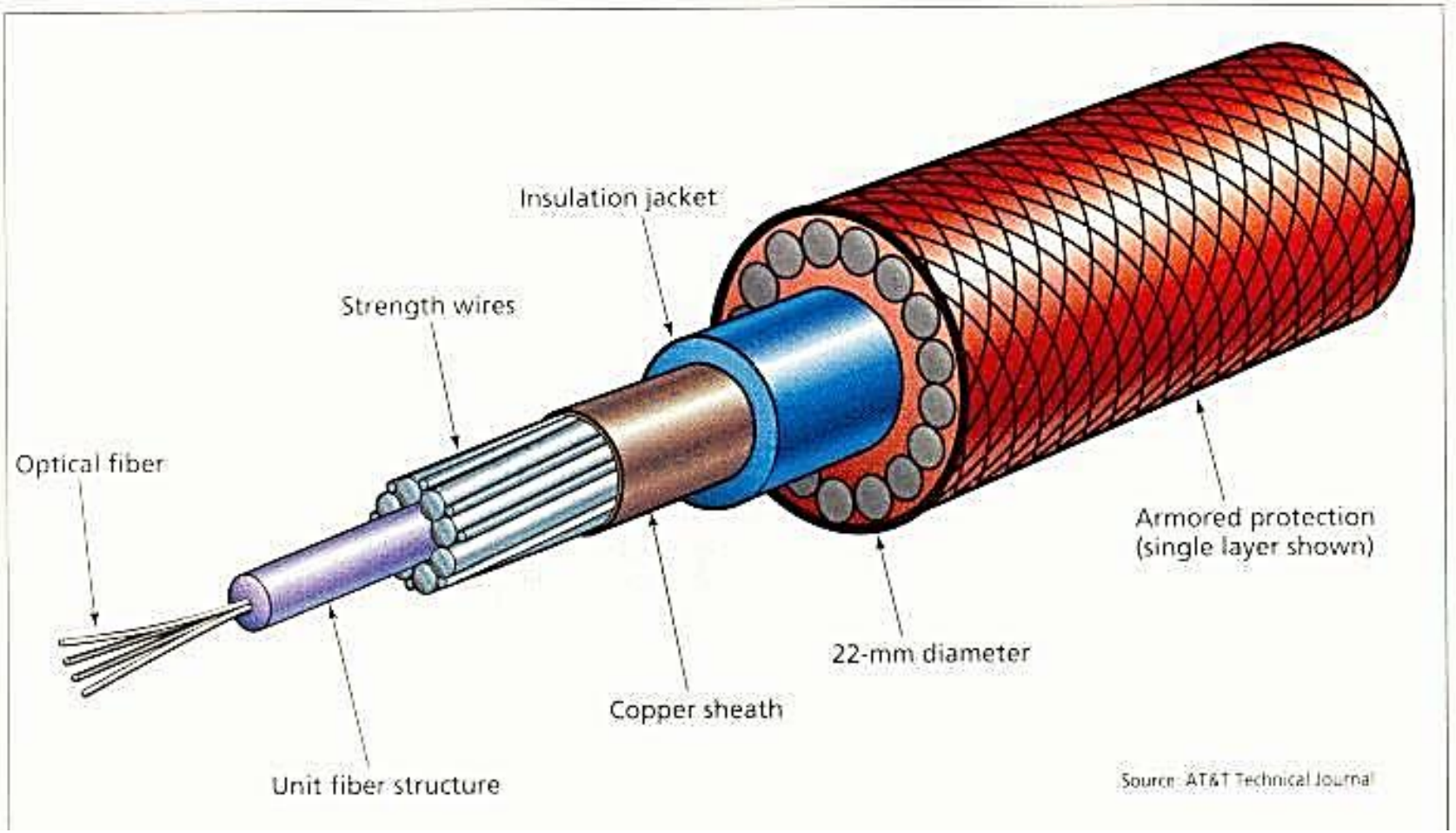


- Telecommunications
- Local Area Networks
- Cable TV
- CCTV
- Optical Fiber Sensors

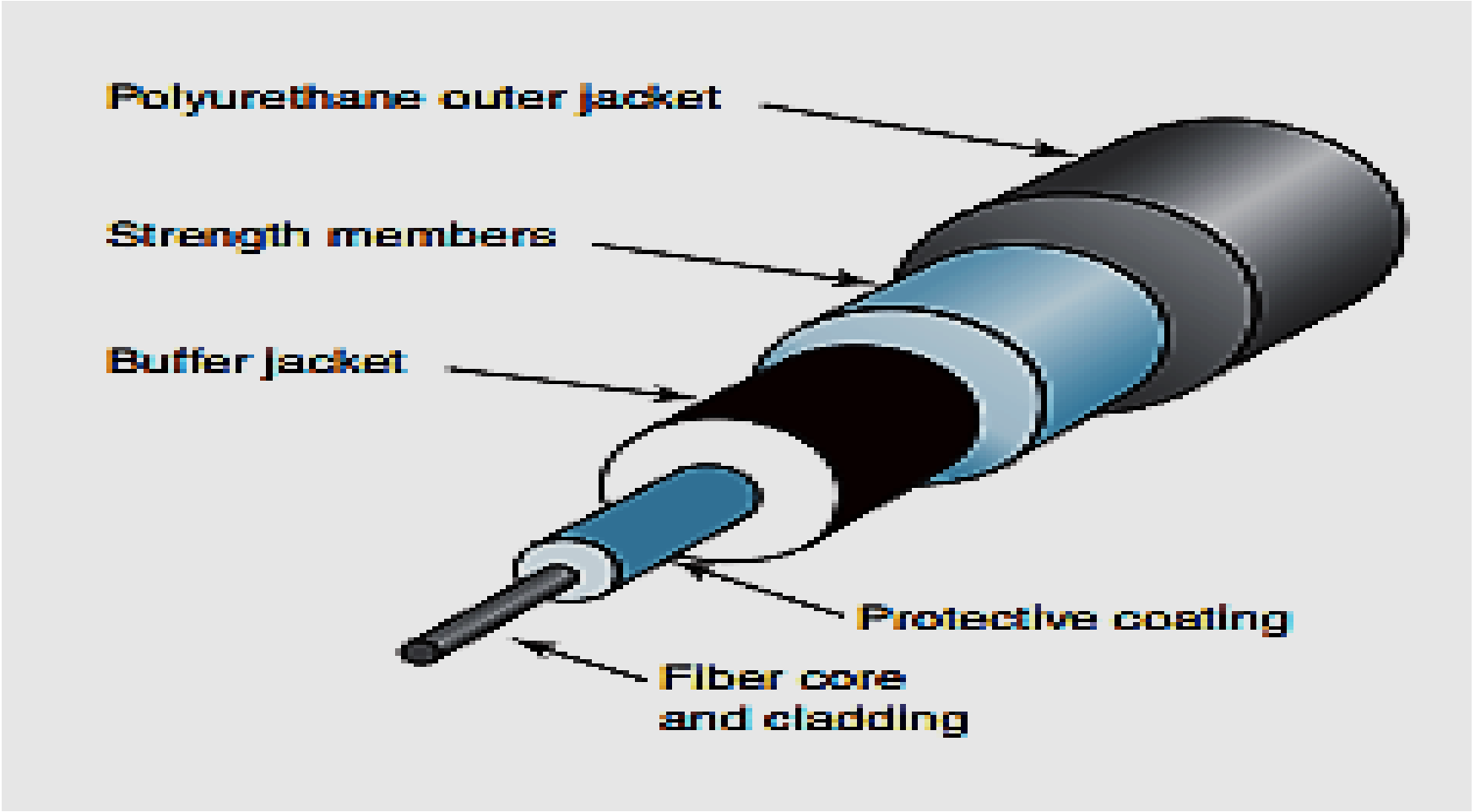
Difference between Optical fiber and Coaxial cable

Key	Optical fiber	Coaxial Cable
Transmission Type	Optical fiber transmits data/signals in the form of light.	The coaxial cable transmits data/signals in the form of electrical signals.
Material	Optical fiber is made using plastic and glass.	Coaxial cable is prepared using plastic and copper wires.
Efficiency	Optical fiber is highly efficient and signal loss is negligible.	Coaxial cable is less efficient.
Cost	Optical fiber is costly and its installation is quite expensive.	Coaxial cable is cheap and its installation is less expensive.
Weight	Optical fiber is quite light in weight.	Coaxial cable is very heavy as compared to an optical fiber.
Diameter	Optical fiber is having a smaller diameter.	Coaxial cable diameter is bigger than that of optical fiber.
Installation	The installation of Optical fiber is complex.	The installation of Coaxial cable is comparatively easy.

Construction of Optical Fiber



Construction of Optical Fiber

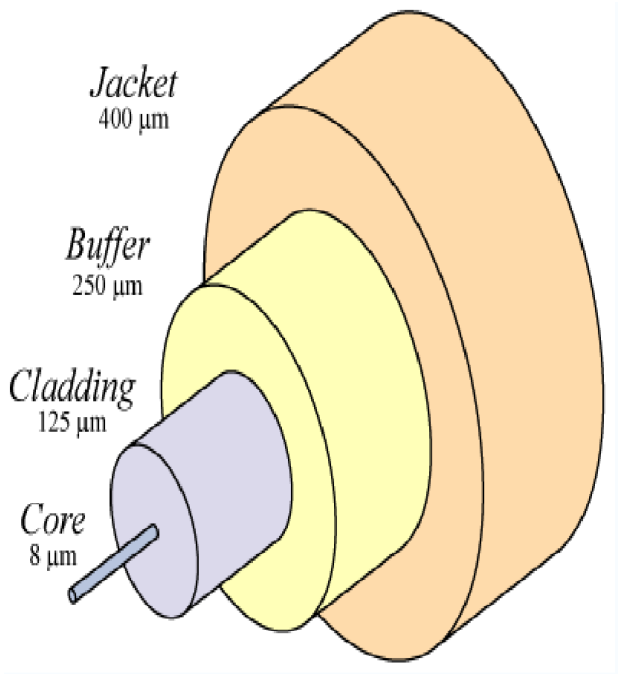


Construction of Optical Fiber

1) **Fiber core and cladding:** The actual fiber portion of an optical cable is generally considered to include both the fiber core and its cladding.
2) **Protective coating:** A special coating is generally applied to the outside of the cladding to seal and preserve the fiber's strength.
3) **Buffer jacket:** The protective coating is surrounded by a buffer jacket which provides the cable additional protection against abrasion and shock.
4) **Strength member:** The buffer jacket is encapsulated in a strength member which increases the tensile strength of the overall cable assembly.
5) **Polyurethane jacket:** The entire cable is protected by a polyurethane jacket.

Construction of Optical Fiber

3 TYPES OF OPTICAL FIBERS



1. Plastic core and cladding
2. Glass core with plastic cladding (called PCS fiber- Plastic Clad Silica)
3. Glass core and glass cladding (called SCS - Silica Clad Silica)

Light Propagation

- Physics of light
  - Photons (light "particles")
    - light represented by tiny bundles of energy (or quanta), following straight line paths along the rays.
  - In free space velocity of electromagnetic wave like light is 3×10^8 m/s
  - Planck's Law

$$E_p = hf$$

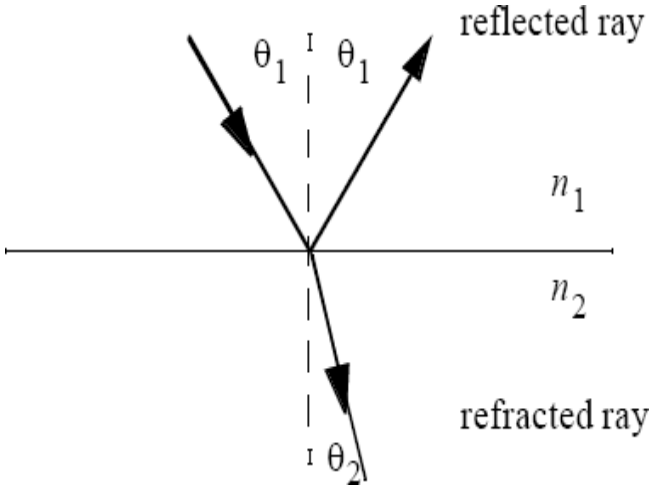
Where,

Ep – energy of the photon (joules)
h = Planck's constant = 6.625 x 10^{-34} J-s
f – frequency o f light (photon) emitted (hertz)

Light Propagation

- Reflection and Refraction of light
  - Snell's Law

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$



where

θ_1 angle of incidence

θ_2 angle of refraction

n_1 refractive index of medium 1

n_2 refractive index of medium 2

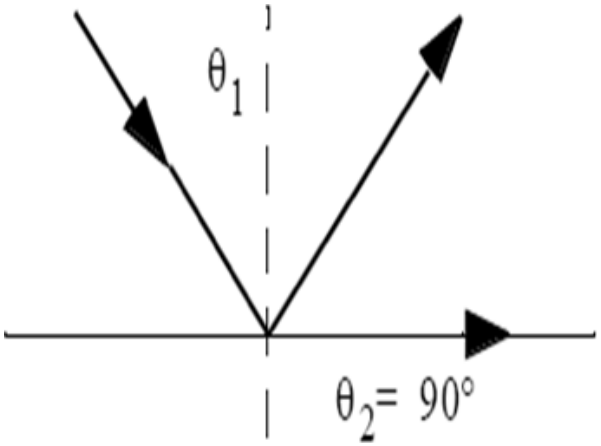
Light Propagation

- Critical Angle
  - The minimum angle of incidence at which the light ray may strike the interface of two media and result in an angle of refraction of 90 ° or greater
  - We must satisfy $n_1 > n_2$

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

$$\sin\theta_1 = \frac{n_2}{n_1}\sin\theta_2 \quad [\text{if } \theta_2 = 90°]$$

$$\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right)$$



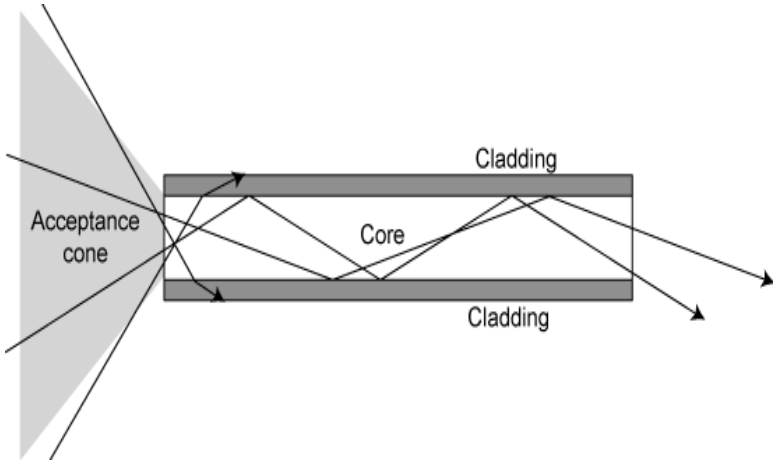
Light Propagation

- Total refraction
  - If the angle of refraction is 90 º or greater,
    - The light ray is not allowed to penetrate the less dense medium
    - Consequently, total reflection takes place
  - Acceptance angle
    - The maximum angle in which external light rays may strike the air/glass interface and still propagate down the fiber.

Light Propagation

- Acceptance angle or acceptance cone

$$\theta_{\text{in(max)}} = \sin^{-1}\left(\frac{\sqrt{n_1^2 - n_2^2}}{n_0}\right)$$



Where,

- n_0 – refractive index of air
- n_1 – refractive index of glass fiber core (1.5)
- n_2 – refractive index of quartz fiber cladding

Light Propagation

- **Numerical aperture (NA)**
  - Closely related to acceptance angle
  - Used to describe the light-gathering or light-collecting ability of an optical fiber.
  - NA is a dimensionless number
  - It characterizes the range of angles over which the system can accept or emit light

Light Propagation

- **Numerical aperture (NA)**
  - For light entering the glass from air, NA is defined as

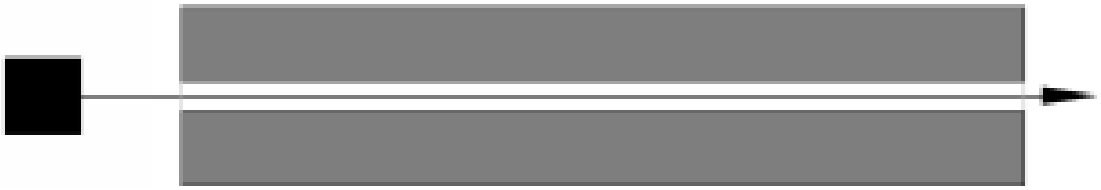
$$NA = \sin\theta_{in}$$
$$= \sqrt{n_1^2 - n_2^2}$$

Mode of Propagation

- Light could propagate either reflection or refraction
- Mode of operation
  - Decides how light propagates
  - It simply means path
- Two main categories of optical fiber
  - Single-mode optical fiber
    - One path of light ray propagates through the fiber
  - Multi-mode optical fiber
    - More than one path of light ray propagates through the fiber

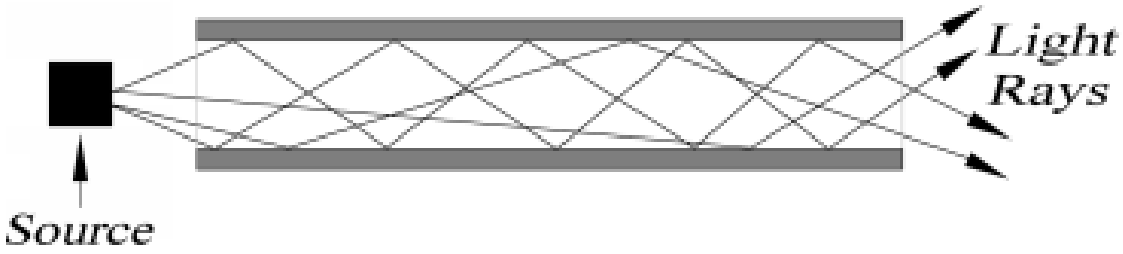
Mode of Propagation

- **Single-mode fibers**
  - Used to transmit one signal per fiber
  - Used in telephone and cable TV.
  - Small cores (9 microns in diameter)
  - Transmit infra-red light from laser.



Mode of Propagation

- **Multi mode fibers**
  - Transmit many signals per fiber
  - Used in computer networks.
  - Larger cores (≥ 50 microns in diameter)
  - Transmit infra-red light from LED.
  - Allowing less precise & cheaper transmitters and receivers to connect to it as well as cheaper connectors.



Mode of Propagation

- Multi mode fibers
  - Multi-mode fiber introduces multimode distortion
    - limits the bandwidth and length of the link.
  - Higher dopant content,
    - more expensive and exhibits higher attenuation.
  - The number of propagating modes possible

$$N = \left\lfloor \left(\frac{\pi d}{\lambda} \sqrt{n_1^2 - n_2^2} \right)^2 \right\rfloor$$

d = core diameter

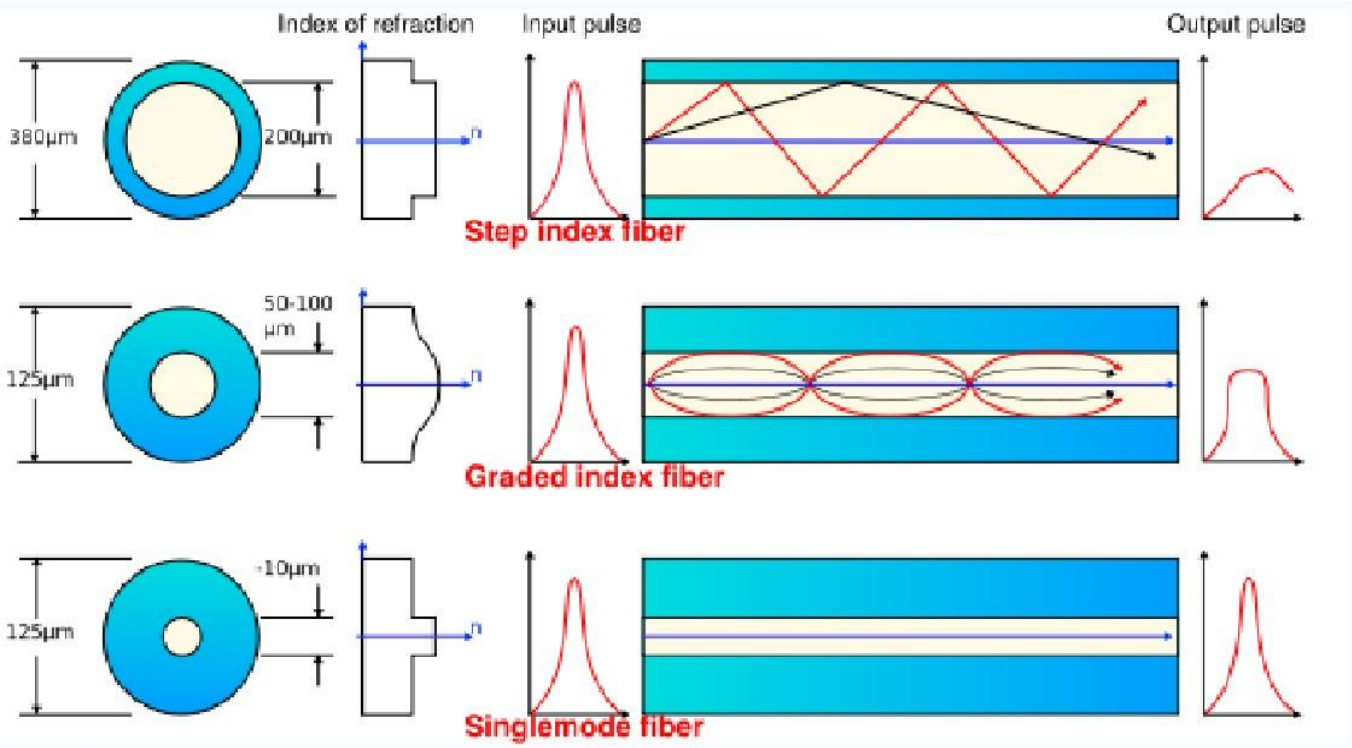
Λ = wavelength

n_1 = reflective index of core

n_2 = reflective index of cladding

Mode of Propagation

- Index profile
  - Graphical representation of magnitude of the reflective index.



Mode of Propagation

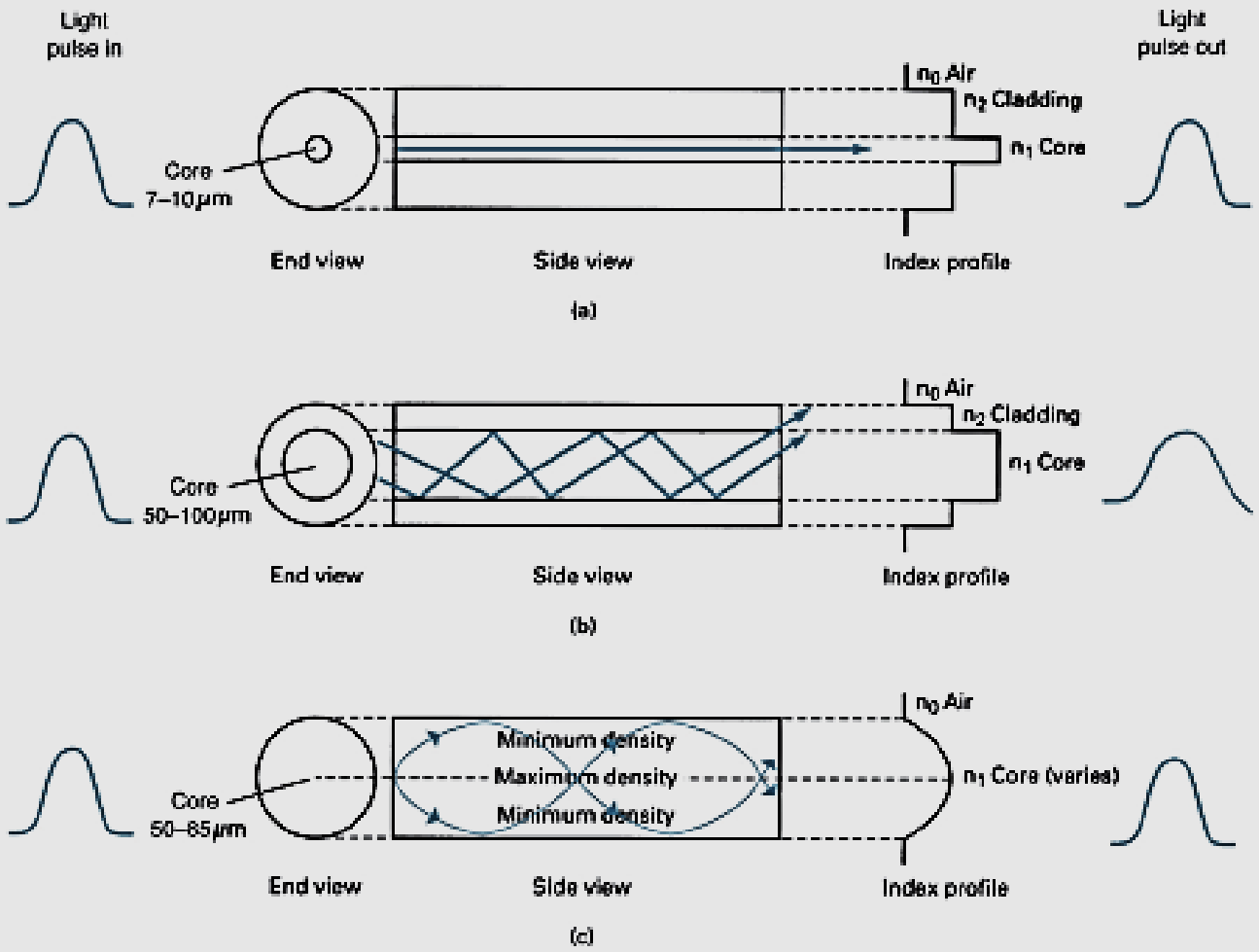


FIGURE 15 Core index profiles: (a) single-mode step index; (b) multimode step index; (c) multimode graded index

Optical Fiber Classification

- Single-mode step-index optical fiber
  - the light is guided down the center of an extremely narrow core
- Multi-mode step-index optical fiber
  - the reflective walls of the fiber move the light pulses to the receiver
- Multi-mode Graded-index optical fiber
  - acts to refract the light toward the center of the fiber by variations in the density
- Comparison of fibers
  - Advantages and disadvantages of each type of fiber-self study

Single-mode step-index fiber

Advantages include the following:

1. Minimum dispersion: All rays propagating down the fiber take approximately the same path; thus, they take approximately the same length of time to travel down the cable. Consequently, a pulse of light entering the cable can be reproduced at the receiving end very accurately.

2. Because of the high accuracy in reproducing transmitted pulses at the receive end, wider bandwidths and higher information transmission rates (bps) are possible with single-mode step-index fibers than with the other types of fibers.

Single-mode step-index fiber

Disadvantages include the following:

1. Because the central core is very small, it is difficult to couple light into and out of this type of fiber. The source-to-fiber aperture is the smallest of all the fiber types.

2.Again, because of the small central core, a highly directive light source, such as a laser, is required to couple light into a single-mode step-index fiber.

3.Single-mode step-index fibers are expensive and difficult to manufacture.

Multimode step-index fiber

Advantages include the following:

1.Multimode step-index fibers are relatively inexpensive and simple to manufacture.

2.It is easier to couple light into and out of multimode step-index fibers because they have a relatively large source-to-fiber aperture.

Disadvantages include the following:

1. Light rays take many different paths down the fiber, which results in large differences in propagation times.

2. The bandwidths and information transfer rates are less other types of fiber cables.

Multimode graded -index fiber

Essentially, there are no outstanding advantages or disadvantages of this type of fiber. Multimode graded-index fibers are easier to couple light into and out of than single-mode step-index fibers but are more difficult than multimode step-index fibers. Distortion due to multiple propagation paths is greater than in single-mode step-index fibers but less than in multimode step-index fibers. This multimode graded-index fiber is considered an intermediate fiber compared to the other fiber types.

Losses in Optical Fiber Cable

- Power loss
  - Most important characteristic of fiber
  - Often call attenuation
  - Diverse effects
    - Reducing the bandwidth and data rate
    - Lower efficiency and overall system capacity
    - Attenuation (unit less) is defined as

$$A_{(dB)} = 10 \log \left(\frac{P_{out}}{P_{in}} \right)$$

P_{in} and P_{out} are cable input and output power

Losses in Optical Fiber Cable

- Fiber cable attenuation

Table 1-3 Fiber Cable Attenuation

Cable Type	Core Diameter (μm)	Cladding Diameter (μm)	NA (unitless)	Attenuation (dB/km)
Single mode	8	125	—	0.5 at 1300 n
	5	125	—	0.4 at 1300 n
Graded index	50	125	0.2	4 at 850 nm
	100	140	0.3	5 at 850 nm
Step index	200	380	0.27	6 at 850 nm
	300	440	0.27	6 at 850 nm
PCS	200	350	0.3	10 at 790 nm
	400	550	0.3	10 at 790 nm
Plastic	—	750	0.5	400 at 650 nr
	—	1000	0.5	400 at 650 nr

Losses in Optical Fiber Cable

- Attenuation vs Distance
  - Optical power measured in a given distance from the source is

$$P = Pt \times 10^{-Al/10}$$

$$P(dBm) = P_{in}(dBm) - Al(dB)$$

Losses in Optical Fiber Cable

– Example-

- For a single mode optical cable with 0.25 dB/km loss determine the optical power 100 km from a 0.1 mW light source.

Pout

Pin = 10 log (P/1 mW)

Loss

$$P(\text{dBm}) = 10\log\left(\frac{0.1\text{ mW}}{0.001\text{ W}}\right) - [(100\text{ km})(0.25\text{ dB/km})]$$

$$= -10\text{ dBm} - 25\text{ dB}$$

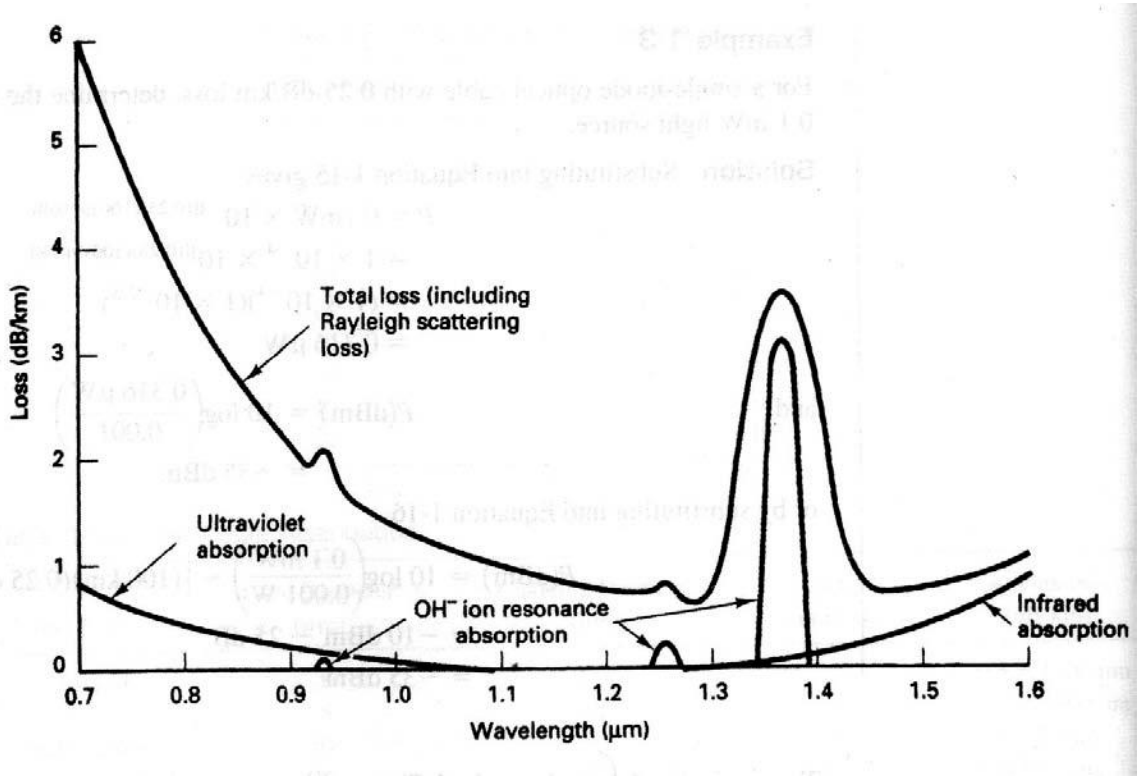
$$= -35\text{ dBm}$$

Losses in Optical Fiber Cable

- Absorption loss
  - Analogues to power dissipation in copper cable
    - Impurities of fiber absorb light and convert it to heat
    - 99.9999% pure glass has loss between 1dB/km and 1000 dB/Km
  - Three factors
    - Ultra violate absorption
      - Caused by valance electrons of silica
      - Light ionizes the valance electron
    - Infrared absorption
      - Absorption of photon of light by the atom of class core molecules
    - Ion resonance absorption
      - Caused by OH^- ions in the material
        - » Trapped in the glass during manufacturing process
      - Iron, copper and chromium molecules also caused ion absorption

Losses in Optical Fiber Cable

- Absorption loss
  - Absorption loss in optical fiber

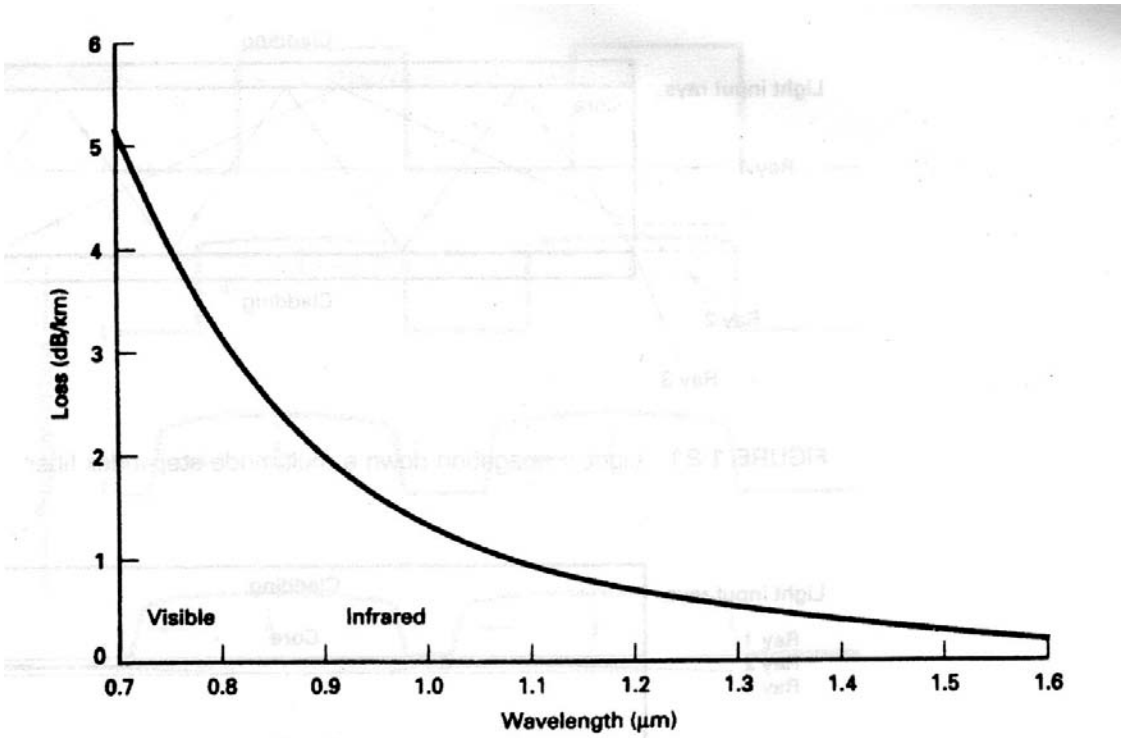


Losses in Optical Fiber Cable

- Material or Rayleigh Scattering Losses
  - Fibers are very long and small diameter
  - In manufacturing process
    - Glass in plastic state (not liquid and not solid)
    - In the cooling process develop permanent submicroscopic irregularities
  - Such irregularities causes escaping some light through cladding
    - Cause a loss in light power and called Rayleigh Scattering loss.

Losses in Optical Fiber Cable

- Material or Rayleigh Scattering Losses
  - Rayleigh Scattering Losses as a function of wavelength



Losses in Optical Fiber Cable

- Chromatic or wavelength dispersion
  - Light sources such as LED contain many wavelengths
  - Light of different wavelength travel in different velocity
  - At a long distance from the transmitter, light of different wavelength arrive at different time.
  - This resulting an impairment called chromatic dispersion.

Losses in Optical Fiber Cable

- Radiation Loss
  - Caused mainly by small bends and kinks
  - Two types of bends: micro-bends and constant radius bends
  - Micro-bends
    - Miniature bends and geometric imperfection along the axis of the fiber
  - Constant radius bends
    - Caused by excessive pressure and tension
    - Generally occur during handling or installation

Losses in Optical Fiber Cable

- Modal Dispersion
  - Sometimes called pulse spreading
  - Different light rays take different paths
    - Caused the differences between the propagation time of light rays
  - Only occurs in multimode fibers
    - Graded index fiber reduced it considerably
  - Cause a pulse of light energy to spread in time
    - One pulse may interfere the other

Losses in Optical Fiber Cable

- **Modal dispersion** is a distortion mechanism occurring in multimode fibers and other waveguides, in which the signal is spread in time because the propagation velocity of the optical signal is not the same for all modes. Other names for this phenomenon include **multimode distortion**, **multimode dispersion**, **modal distortion**, **intermodal distortion**, **intermodal dispersion**, and **intermodal delay distortion**.
- Modal dispersion should not be confused with chromatic dispersion, a distortion that results due to the differences in propagation velocity of different wavelengths of light. Modal dispersion occurs even with an ideal, monochromatic light source.

Losses in Optical Fiber Cable

- Modal Dispersion
  - Light propagation down different fibers

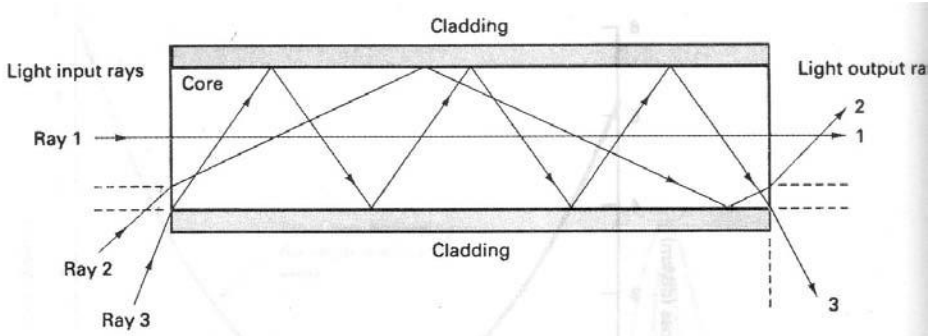


FIGURE 1-21 Light propagation down a multimode step-index fiber

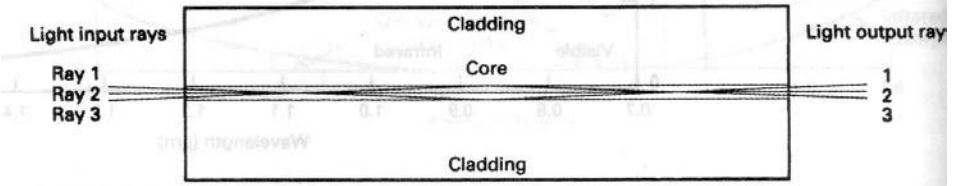


FIGURE 1-22 Light propagation down a single-mode step-index fiber

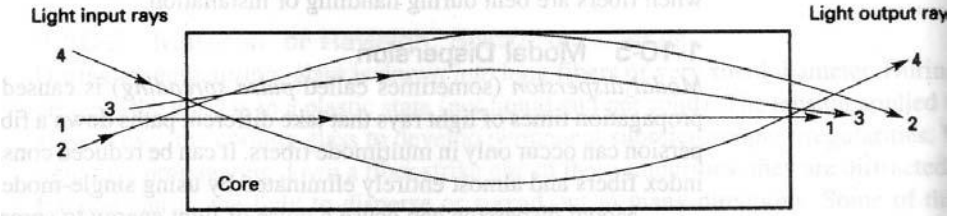
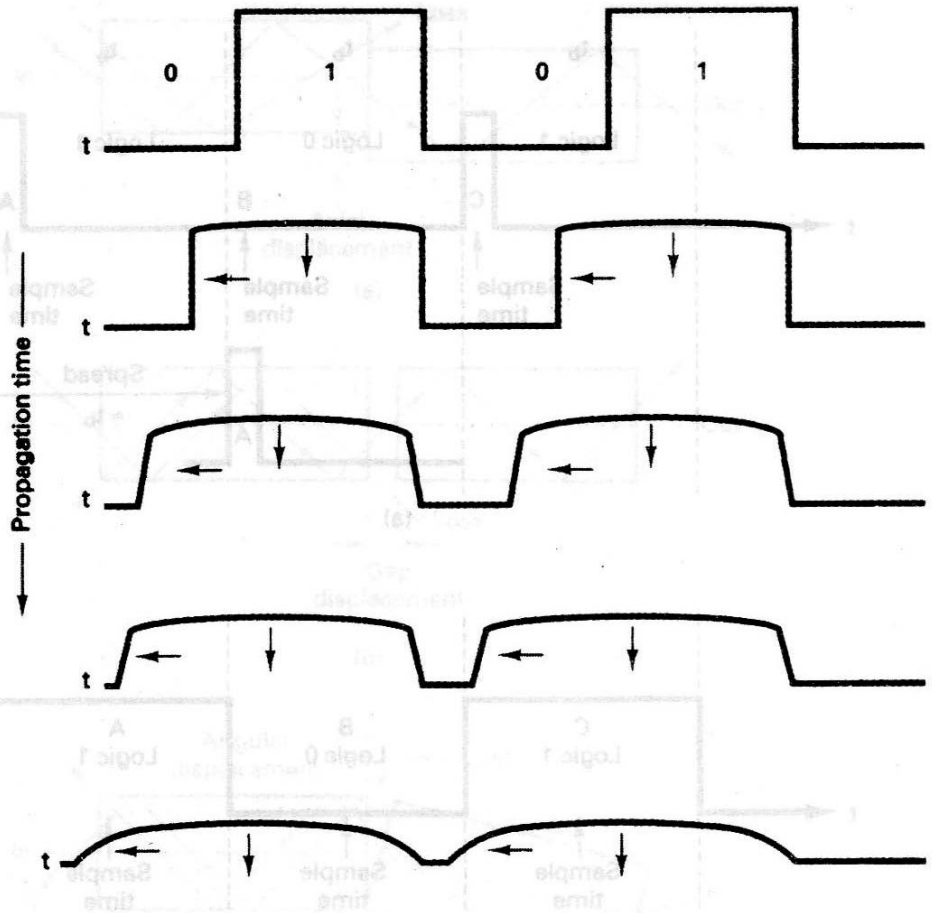


FIGURE 1-23 Light propagation down a multimode graded-index fiber

Losses in Optical Fiber Cable

- Modal Dispersion
  - Pulse width dispersion in an optical fiber

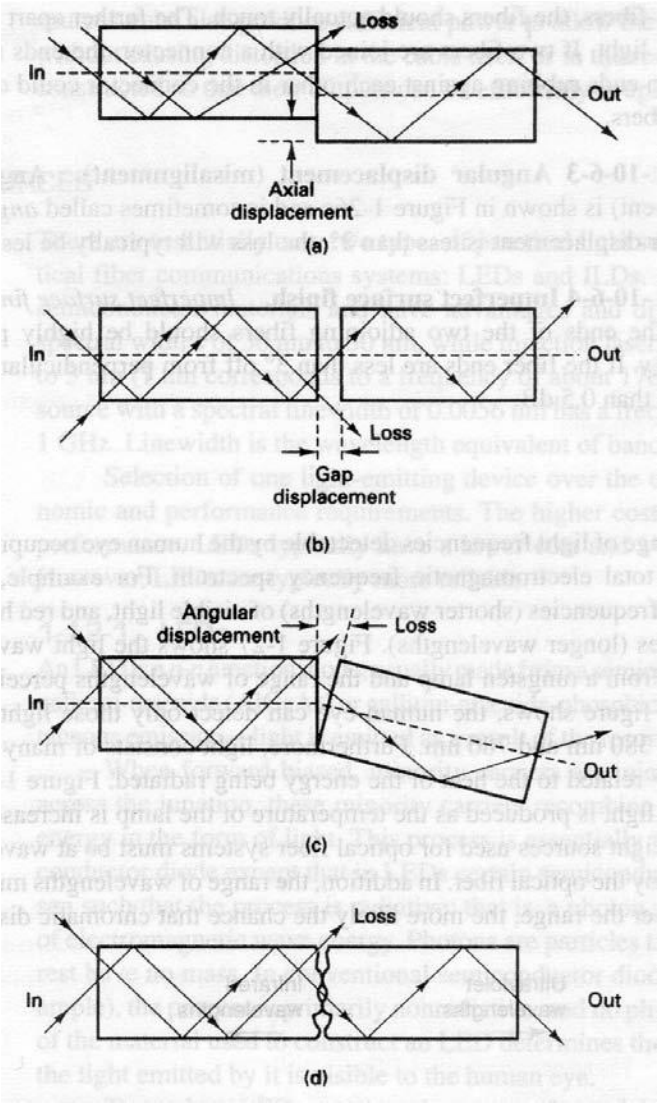


Losses in Optical Fiber Cable

- Coupling Losses
  - Caused by imperfect physical contacts in various junctions
    - Light source to fiber connections
    - Fiber to fiber connections
    - Fiber to photo detector connections
  - Mostly junction loss is caused by alignment problems

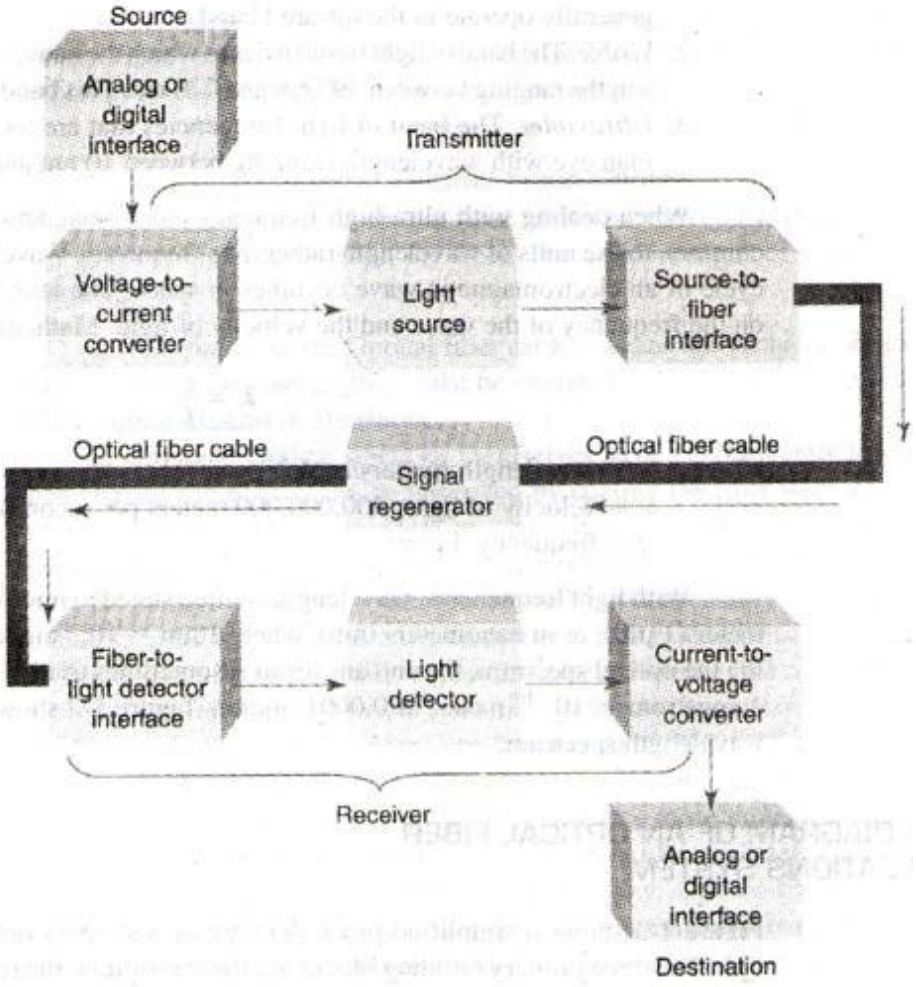
Losses in Optical Fiber Cable

- Coupling Losses
  - Lateral displacement
  - Gap-displacement
  - Angular displacement
  - Imperfect surface finish



Optical Fiber Communication link

- Block Diagram



Optical Sources

- Tow types of practical light sources
  - LED (Light Emitting Diode)
    - Spectral width 30 nm to 50 nm
  - ILD (Injection Laser Diode)
    - Spectral width 1 nm to 3 nm

Optical Sources

- LED (Light Emitting Diode)
  - PN junction diode
    - Made of AlGaAs, GaAsP, etc
  - Emit light by spontaneous emission
    - As a result of recombination of electrons and holes
  - Minority carriers are injected across the PN junction
    - Minority carrier recombines with majority carrier
    - Give up energy in the form of light.
  - Conventional Diode vs LED
    - Material and dopants are chosen for LEDs so that photons are produced.
    - Photons with no mass travels at the speed of light

Optical Sources

- LEDs
  - Common semiconductor materials used in LED construction and their respective output wavelength

Table 1-4 Semiconductor Material Wavelengths

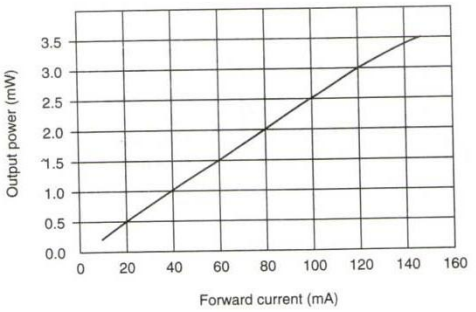
Material	Wavelength (nm)
AlGaInP	630–680
GaInP	670
GaAlAs	620–895
GaAs	904
InGaAs	980
InGaAsP	1100–1650
InGaAsSb	1700–4400

Optical Sources

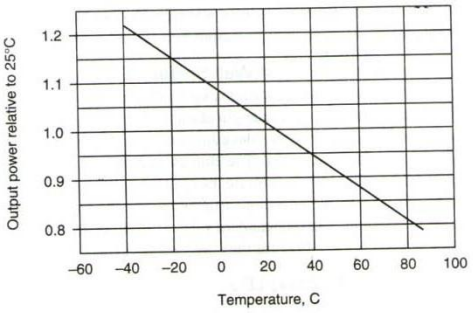
- LEDs
  - Homojunction LEDs
    - PN junction made from two different mixtures of the same type of atom.
    - Also known as surface emitters (light emits from the surface of the LED)
    - Output approximately 500 µW at a wavelength of 900 nm
    - Non directionality of light emission
      - Poor choice for optical fiber system
  - Heterojunction LEDs
    - Different type of atoms are used for P nad N type semiconductor meterials
    - Also known as edge emitters (light emits from the edge of the LED)

Optical Sources

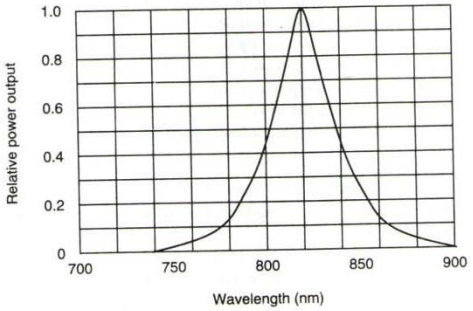
- LEDs
  - Heterojunction LEDs
    - Different type of atoms are used for P nad N type semiconductor meterials
    - Also known as edge emitters (light emits from the edge of the LED)
    - Characteristics are shown in fig.



(a)



(b)



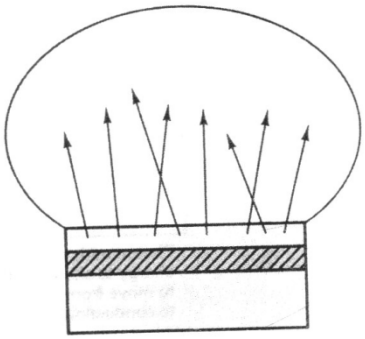
(c)

Optical Sources

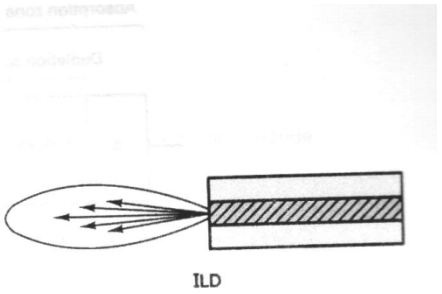
- LEDs
  - Burrus Etched-Well Surface-Emitting LED
    - Developed by Burrus and Dawson of bell Laboratory
    - Designed for practical application such as telecommunication.
    - Surface emitting LED which emits light in many directions
    - Etched –wall helps concentrate the light in small area
    - Allows more power to be coupled in to the optical fiber.
    - Difficult and expensive to manufacture
  - Eadge emitting LED
    - Emits a more directional light pattern than surface emitting LEDs but less amount of light
    - Coupling loss is less than surface emitting LEDs

Optical Sources

- ILDs
  - Lasers are constructed from gases, glass, liquid, solid, etc.
    - For optical fiber communications application lasers are made from semiconductor
  - ILD is similar to LED below a certain threshold current.
  - ILD can be used at higher bit rate than LEDs
  - ILDs generate monochromatic light
    - Reduces chromatic or wavelength dispersion
  - ILDs are 10 times more expensive than LEDs
  - Operates at high power
    - Less lifetime than LEDs
  - More temperature dependent



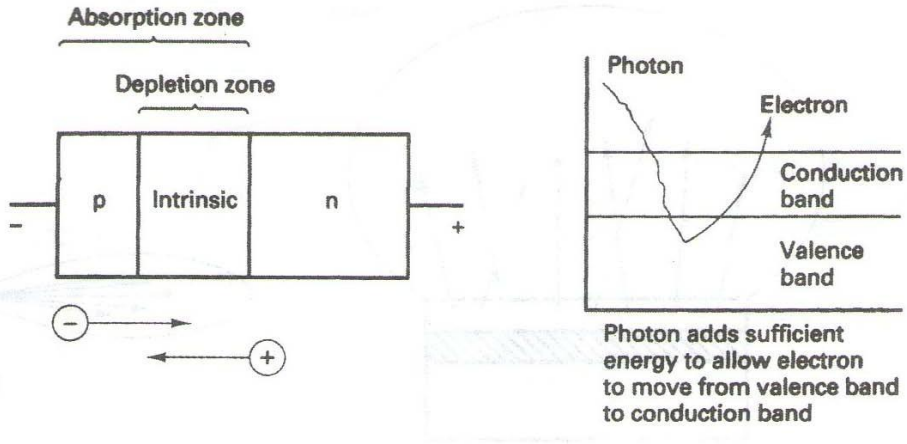
LED



ILD

Light Detectors

- PIN diode
  - Depletion layer photodiode
  - Most common device to detect light in Optical Fiber communication system
  - The intrinsic “I" region is flooded with charge carriers from the “P" and “N" regions
  - PIN photo diode operates just the opposite of the LEDs
    - Most of the photons are absorbed by the electrons of balance band of the intense material
    - Absorbed photons generates sufficient energy for current flow



Light Detectors

- APD (Avalanche photo diode)
  - pipn structure
    - Light is absorped by the thin havey doped n-layer
    - High electric field intensity developed across the i-p-n junction by reverse bias causes impact ionization to occur
    - During impact ionization carrier gains sufficient energy to ionize other bound electrons
      - This ionization cause more ionization to occur and the process continues till an avalanche
    - APDs are more sensitive than PIN diods
      - Less amplification is required
    - APDs major disadvantage is long transit time.

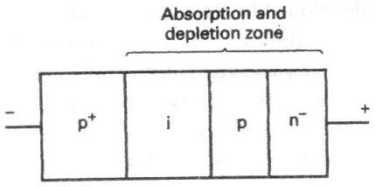
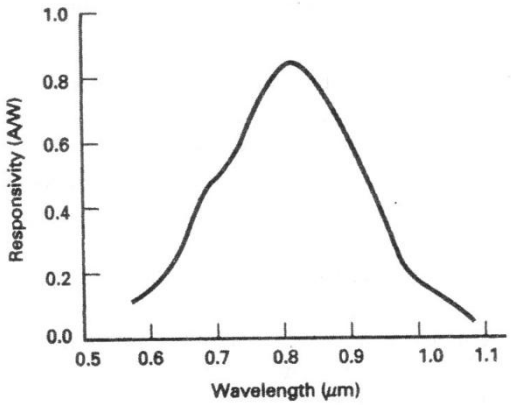


FIGURE 1-38 Avalanche photo-diode construction

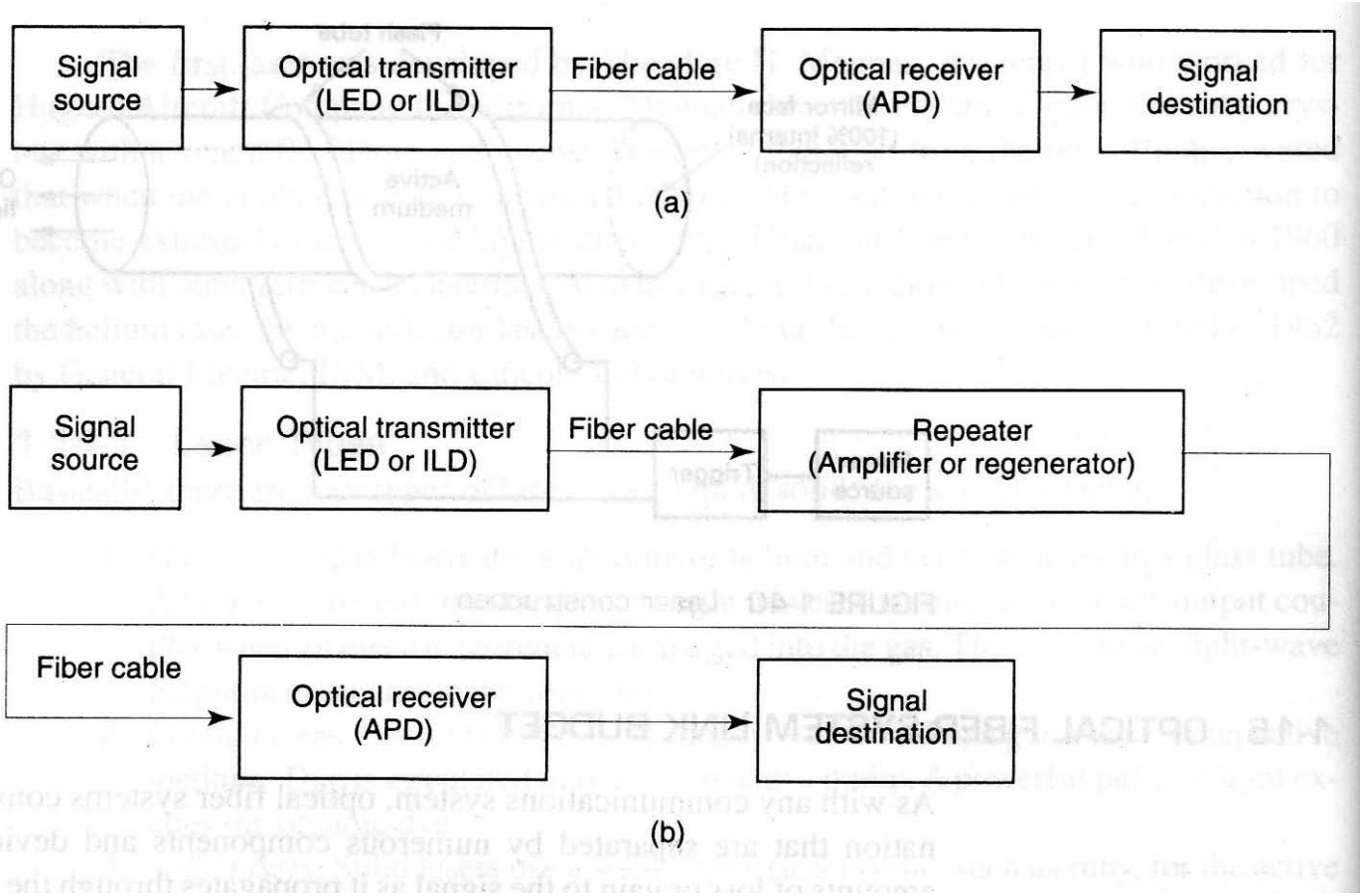
Light Detectors

- Characteristics of Light Detectors
  - Responsively
    - Measure of conversion efficiency
    - Ratio of output current to the input optical power
    - Dependent on the wavelength or frequency
  - Dark Current
    - The leakage current with no light input
    - Thermally generated carrier cause this
  - Transit time
    - The time it takes a light-induced carrier to travel across the depletion region
    - Parameter dettermines the maximum bit rate
  - Spectral Response
    - The range of wavelength values that a photodiode will respond.
  - Light Sensitivity
    - The minimum optical power fro which a light detector can reproduce effective electrical signal.



Optical fiber system Link Budget

- Optical fiber system



Optical fiber system Link Budget

- The link Budget
  - Light power sources
  - Light detector
  - Various cable and connector losses
    - Cable losses
    - Connector losses
    - Source to cable interface loss
    - Cable to light detector interface loss
    - Splicing loss
    - Cable bend loss

Optical fiber system Link Budget

- Example
  - Determine the optical power received in dBm and watt for a 20 km optical fiber link with the following parameters
    - LED output power of 30 mW
    - Four 5 km section of optical cable with a loss of 0.5 dB/km
    - Three cable-to-cable connector with a loss of 2 dB each
    - Source to cable interface loss of 0.9 dB
    - Cable to light detector interface loss of 2.1 dB
    - No loss due to the cable bend

Solution The LED output power is converted to dBm using Equation 6:

$$P_{out} - 10 \log \frac{30 \text{ mW}}{1 \text{ mW}}$$

$$= 14.8 \text{ dBm}$$

The cable loss is simply the product of the total cable length in km and the loss in dB/km. Four 5-km sections of cable is a total cable length of 20 km; therefore,

$$\text{total cable loss} = 20 \text{ km} \times 0.5 \text{ dB/km}$$

$$= 10 \text{ dB}$$

Cable connector loss is simply the product of the loss in dB per connector and the number of connectors. The maximum number of connectors is always one less than the number of sections of cable. Four sections of cable would then require three connectors; therefore,

$$\text{total connector loss} = 3 \text{ connectors} \times 2 \text{ dB/connector}$$

$$= 6 \text{ dB}$$

The light source-to-cable and cable-to-light detector losses were given as 1.9 dB and 2.1 dB, respectively. Therefore,

$$\text{total loss} = \text{cable loss} + \text{connector loss} + \text{light source-to-cable loss} + \text{cable-to-light detector loss}$$

$$= 10 \text{ dB} + 6 \text{ dB} + 1.9 \text{ dB} + 2.1 \text{ dB}$$

$$= 20 \text{ dB}$$

The receive power is determined by substituting into Equation 22:

$$P_r = 14.8 \text{ dBm} - 20 \text{ dB}$$

$$= -5.2 \text{ dBm}$$

$$= 0.302 \text{ mW}$$